The Emergence of the Coulomb Force and Orbital Stability from Time-Density Gradients in Super Information Theory

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Abstract

Standard Quantum Mechanics treats the Hydrogen atom as a probabilistic solution to the Schrödinger equation within a postulated electrostatic potential $V(r) \propto -1/r$. Super Information Theory (SIT) offers a fundamental derivation of this structure based on the Principle of Informational Realism. We posit that the atomic nucleus is not merely a source of charge, but a topological defect of high coherence that dilates the local Time-Density field ρ_t. By applying the SIT Master Energy Equation, $E(r) = \zeta \cdot R_{\text{coh}} \cdot [\rho_t(r)]^2$, to a radial gradient of the form $\rho_t(r) \approx \rho_0(1 - \kappa/r)$, we demonstrate that the gradient of the squared time-density field yields an effective entropic force $F \propto -1/r^2$. This recovers the attractive inverse-square law of the Coulomb interaction as an emergent geometric effect. Furthermore, we show that the quantization of electron shells is enforced by the topological stability of the Coherence Field ψ. For the electron to sustain a non-zero mass ($R_{\text{coh}} > 0$), the holonomy of its orbital trajectory must be Identity ($\oint \nabla\theta \cdot dl = 2\pi n$), ensuring single-valued phase history and constructive interference. Thus, the Rydberg energy levels are derived not as arbitrary eigenvalues, but as the discrete geometric zones where informational coherence is conserved against the vacuum.

1 Introduction: From Probability to Informational Topology

The stability of matter remains one of the most profound inquiries in physics. In the Standard Model, the Hydrogen atom is described as a probabilistic system where the electron occupies a standing wave probability cloud, defined as an eigenstate of the Schrödinger equation within a postulated electrostatic potential $V(r) \propto 1/r$. While this framework is predictively accurate, it is ontologically silent on the origin of the potential itself or why the electron's mass persists as a stable quantity.

Super Information Theory (SIT) reframes this problem through the lens of *Informational Realism*. We posit that the fundamental substrate of reality is not a pre-existing manifold of mass and energy, but a dynamic interaction between two scalar fields: the Coherence Ratio (R_{coh}), representing the density of order, and the Time-Density (ρ_t), representing the rate of causal updates.

In this framework, the electron is not a fundamental particle but an emergent, localized excitation of the coherence field. Its stability is not governed by arbitrary quantum numbers, but by the topological conservation of information. This paper derives the structure of the Hydrogen atom from the SIT Master Energy Equation, demonstrating that the Coulomb force emerges naturally from the gradient of the time-density field, and that orbital quantization is a geometric necessity for maintaining unitary holonomy.

1.1 The Emergence of Mass and Force

Standard physics assumes mass (m) is an intrinsic property. SIT derives mass as a consequence of informational density. The energy density of any region is given by the Master Energy Equation:

$$E(x) = \zeta \cdot R_{\text{coh}}(x) \cdot [\rho_t(x)]^2 \tag{1}$$

where ζ is the Informational Inertia Constant. Consequently, the rest mass of a particle is defined as the localized storage of this informational energy:

$$m(x) = \frac{\zeta}{c^2} R_{\text{coh}}(x)[\rho_t(x)]^2 \tag{2}$$

By treating the atomic nucleus as a source of high coherence that distorts the local time-density field ρ_t, we show that the gradient of the squared field, $\nabla(\rho_t^2)$, naturally reproduces the $1/r^2$ behavior of the electric force without invoking charge as a distinct primitive.

1.2 Stability via Identity Holonomy

Why are electron orbits quantized? In SIT, an electron exists only where the coherence field can maintain a non-zero amplitude ($R_{\text{coh}} > 0$). This requires that the phase history of the particle along its closed trajectory γ allows for constructive self-interference. This condition is formally expressed through the Holonomy (or Wilson Loop) of the coherence bundle:

$$U[\gamma] = \exp\left(i\frac{q}{\hbar}\oint_\gamma A_\mu dx^\mu\right) \tag{3}$$

For the particle to avoid decoherence—a state where $R_{\text{coh}} \to 0$ and mass vanishes—the holonomy must be Identity ($U[\gamma] = 1$). This forces the phase winding number to be an integer:

$$\oint_\gamma \nabla\theta \cdot dl = 2\pi n \tag{4}$$

Thus, the discrete shells of the Hydrogen atom ($n = 1, 2, 3...$) are derived as the distinct topological zones where the coherence field creates a single-valued, self-reinforcing resonance within the nuclear time-density gradient.

2 Theoretical Framework: The SIT Master Equations

Super Information Theory builds upon the foundational postulate of *Informational Realism*: the fundamental substrate of reality is not matter or spacetime, but Information, quantified by a physical field representing Coherence. Observable phenomena, including mass and force, are manifestations of the dynamics of this substrate.

To formalize this, we define the *Two-Field Basis* of Coherence. These two coupled, gauge-invariant scalar fields are defined at every point in spacetime and serve as the primitives for our derivation.

2.1 The Two-Field Basis

- **The Coherence Ratio (R_{coh}):** This is a dimensionless scalar field ($0 \leq R_{\text{coh}} \leq 1$) representing the normalized mutual information density of a spacetime region. It quantifies the degree of local phase alignment and structured correlation, serving effectively as the physical measure of *order*.

- **The Time-Density (ρ_t):** This is a scalar field with units of $[T^{-1}]$ representing the local rate of informational state transitions. It governs the flow of proper time and the effective energy scale of interactions, serving as the physical measure of *dynamics*.

2.2 The Principle of Informational Energy Equivalence

In standard physics, energy is often treated as a property of mass ($E = mc^2$). SIT inverts this relationship through the *Principle of Informational Energy Equivalence*. We posit that energy is fundamentally an emergent property derived from the system's Coherence and its local Time-Density.

This relationship is captured in the SIT Master Energy-Density Equation [1]:

$$E(x) = \zeta \cdot R_{\text{coh}}(x) \cdot [\rho_t(x)]^2 \tag{5}$$

Where:

- $E(x)$ is the Informational Energy Density at spacetime point x.
- ζ (zeta) is the *Informational Inertia Constant*. Dimensional analysis requires this fundamental constant to carry units of $[ML^2]$ (mass $\cdot$ length squared), physically analogous to a *moment of inertia*, relating informational updates to physical resistance.

Conceptually, this equation replaces the constant speed of light c in Einstein's formulation with the dynamic local Time-Density ρ_t. It implies that the energy of a system scales linearly with its internal order (R_{coh}) and quadratically with the rate of its causal updates (ρ_t^2).

2.3 Emergent Mass

A central consequence of this framework is that mass is not an intrinsic property of matter. Instead, "mass" is simply the name given to stable, localized informational energy. By equating the standard energy density with the SIT energy density, we derive the definition for emergent rest mass:

$$m(x) = \frac{\zeta}{c^2} R_{\text{coh}}(x)[\rho_t(x)]^2 \tag{6}$$

In this view, a "massive" particle such as an electron is a stable excitation of the SIT fields characterized by a non-zero intrinsic R_{coh} and a stable ρ_t profile. Conversely, "massless" particles are propagating waves where the effective rest-frame coherence approaches zero. This definition is critical for the following section, where we analyze how such a massive excitation behaves within a gradient of the time-density field.

3 The Emergence of the Coulomb Force

To derive the structure of the Hydrogen atom, we must first establish the nature of the force that binds the electron to the nucleus. In the Standard Model, this is the electrostatic Coulomb force, $F = -k_e q_1 q_2 / r^2$. In Super Information Theory, we do not posit "charge" as a fundamental primitive. Instead, we treat the interaction as an emergent geometric effect arising from the gradient of the Time-Density field.

3.1 The Nucleus as a Time-Density Well

We model the atomic nucleus (proton) not as a point charge, but as a topological defect or "knot" of high coherence. This defect creates a distortion in the local vacuum, creating a "well" in the Time-Density field ρ_t (analogous to gravitational time dilation, where the rate of causal updates slows near a source).

Consistent with the behavior of scalar field potentials, we model the radial profile of ρ_t as a perturbation of the vacuum density ρ_0:

$$\rho_t(r) \approx \rho_0 \left(1 - \frac{\kappa}{r}\right) \tag{7}$$

where κ represents the characteristic coherence length of the nucleus.

3.2 Derivation of the Attractive Inverse-Square Law

We now apply the SIT Master Energy-Density Equation to this radial profile to determine the effective energy landscape experienced by the electron.

Substituting Equation (3) into Equation (1):

$$E(r) = \zeta \cdot R_{\text{coh}} \cdot \left[\rho_0 \left(1 - \frac{\kappa}{r}\right)\right]^2 \tag{8}$$

Expanding the square term:

$$[\rho_t(r)]^2 = \rho_0^2 \left(1 - \frac{2\kappa}{r} + \frac{\kappa^2}{r^2}\right) \tag{9}$$

At atomic scales, where the orbital radius r is significantly larger than the nuclear coherence length κ ($r \gg \kappa$), the second-order term $\frac{\kappa^2}{r^2}$ becomes negligible. The energy density simplifies to:

$$E(r) \approx \zeta R_{\text{coh}} \rho_0^2 \left(1 - \frac{2\kappa}{r}\right) \tag{10}$$

An effective entropic force arises from the gradient of this informational energy landscape. The force $\vec{F}$ is defined as the negative gradient of the energy field, $\vec{F} = -\nabla E(r)$.

Differentiating Equation (6) with respect to r:

$$\begin{aligned} F_{\text{SIT}} &= -\frac{d}{dr}\left(\zeta R_{\text{coh}} \rho_0^2 - \frac{2\zeta R_{\text{coh}} \rho_0^2 \kappa}{r}\right) \\ &= -\zeta R_{\text{coh}} \rho_0^2 \left(\frac{2\kappa}{r^2}\right) \end{aligned} \tag{11}$$

$$F_{\text{SIT}} \propto -\frac{1}{r^2} \tag{12}$$

This result is significant. The negative sign indicates an **attractive** force. It demonstrates that the inverse-square scaling law ($1/r^2$)—the hallmark of the Coulomb interaction—emerges naturally from the gradient of the squared Time-Density field. The electron falls into the "Time-Density Well" to minimize its informational energy.

3.3 Interpretation

In this framework, the "electric field" is reinterpreted as the spatial gradient of the causal update rate ($\nabla\rho_t$). The electron does not orbit because of an intrinsic charge responding to a photon exchange; it orbits because it resides in a steep informational gradient where the energy density falls off as $1/r^2$. This geometric derivation satisfies the necessary condition for stable orbital mechanics without invoking the concept of electric charge as a primary ontological entity.

4 Topological Stability and Orbital Quantization

Having established that the time-density gradient produces an attractive inverse-square force, we must now address why the electron does not radiate its energy and spiral into the nucleus, as classical electrodynamics would predict. In Super Information Theory, this stability is not a postulate but a requirement of *Coherence Conservation.*

4.1 Coherence Conservation

Recall the SIT definition of emergent mass: $m(x) \propto R_{\text{coh}}(x)[\rho_t(x)]^2$. For an electron to exist as a massive particle, it must maintain a non-zero Coherence Ratio ($R_{\text{coh}} > 0$). If the phase of the coherence field ψ becomes randomized (decoherent), the local expectation value of R_{coh} tends toward zero, and the particle's effective mass vanishes. Thus, stable matter can only exist along trajectories where the coherence field maintains constructive self-interference.

4.2 Holonomy of the Coherence Field

The phase stability of the electron along a closed orbital trajectory γ is governed by the gauge connection of the coherence bundle. This is formally described by the Holonomy (or Wilson Loop) $U[\gamma]$:

$$U[\gamma] = \exp\left(i\frac{q}{\hbar}\oint_\gamma A_\mu dx^\mu\right) \tag{13}$$

In the SIT framework, A_μ represents the connection on the coherence-phase bundle. As the electron traverses its orbit, it accumulates a geometric phase $\Phi = \oint \nabla\theta \cdot dl$.

4.3 The Identity Constraint

For the electron to form a stable standing wave—a resonance in the coherence field—the wavefunction must be single-valued. If the phase history does not return to its origin (modulo 2π), the wave will destructively interfere with itself over multiple orbits, driving $R_{\text{coh}} \to 0$.

Therefore, the condition for existence is that the Holonomy must be Identity ($U[\gamma] = 1$). This imposes a topological constraint on the phase winding number n:

$$\oint_\gamma \nabla\theta \cdot dl = 2\pi n, \quad \text{where } n \in \mathbb{Z}^+ \tag{14}$$

This is the mechanism of *Coherence Gating* applied to orbital mechanics: the electron is physically permitted to possess mass only in zones where this topological condition is met.

4.4 Derivation of the Rydberg Levels

We can now derive the quantization of the orbital radii. Relating the phase gradient to the canonical momentum p via the de Broglie relation ($p = \hbar\nabla\theta$):

$$\oint_\gamma p \cdot dl = nh \tag{15}$$

For a circular orbit of radius r, this yields the quantization of angular momentum:

$$L = mvr = n\hbar \tag{16}$$

Combining this with the SIT force law derived in Section 3 ($F_{\text{SIT}} = \frac{k_{\text{eff}}}{r^2}$), we solve the force balance equation $mv^2/r = k_{\text{eff}}/r^2$:

$$v = \sqrt{\frac{k_{\text{eff}}}{mr}} \tag{17}$$

Substituting v into the angular momentum constraint:

$$m\sqrt{\frac{k_{\text{eff}}}{mr}}r = n\hbar \implies \sqrt{mk_{\text{eff}}r} = n\hbar \tag{18}$$

Solving for r:

$$r_n = \frac{n^2\hbar^2}{mk_{\text{eff}}} \tag{19}$$

This result mirrors the Bohr radius formula. It demonstrates that the discrete electron shells ($n = 1, 2, 3...$) are not arbitrary quantum eigenvalues, but the unique geometric radii where the Coherence Field maintains a unitary holonomy within the nuclear Time-Density gradient.

5 Discussion

The derivation presented above offers a fundamental reinterpretation of atomic physics. While the mathematical predictions for the orbital radii r_n align with the Bohr model and, by extension, the principal quantum numbers of the Schrödinger equation, the ontological mechanism is distinct. We now address the implications of this shift from probabilistic eigenvalues to topological stability.

5.1 From Eigenvalues to Topological Necessity

In standard Quantum Mechanics, the discrete energy levels of the Hydrogen atom arise as eigenvalues of a differential operator (the Hamiltonian) acting on a Hilbert space. The quantization is a result of boundary conditions imposed on the probability amplitude at infinity and the origin.

In Super Information Theory, quantization is not an analytic consequence of a differential equation, but a *topological necessity*. The electron shells are the only geometric zones within the time-density gradient where the informational content of the particle is self-consistent. The integer n is identified as a topological winding number (or Pontryagin index) of the Coherence Field's phase. This suggests that "quantum numbers" are physically essentially measures of topological knotting in the underlying informational substrate.

5.2 The Physics of Forbidden Zones

A unique consequence of the SIT Master Energy Equation is the physical interpretation of the "forbidden" regions between electron shells. In standard physics, the electron simply has zero probability of being observed in these zones (for non-transient states).

In SIT, the exclusion is dynamical. If an electron were placed at a radius r where the holonomy condition fails ($\oint \nabla\theta \cdot dl \neq 2\pi n$), the phase of its coherence field would destructively interfere with itself upon every rotation. This interference drives the local Coherence Ratio R_{coh} toward zero. Since emergent mass is defined as $m \propto R_{\text{coh}}\rho_t^2$, an electron in a forbidden zone effectively loses its mass. It ceases to exist as a massive excitation, dissolving into the vacuum radiation field until it can re-resonate at a stable topological radius. This provides a mechanism for the "quantum jump": it is a transition between the only two manifolds where the particle possesses non-zero mass.

5.3 Macroscopic Implications: The Coherence-Gravity Effect

The relationship $E = \zeta \cdot R_{\text{coh}} \cdot \rho_t^2$ extends beyond the atomic scale. It implies that gravity—which couples to stress-energy—must couple to Coherence.

This leads to the "Smoking Gun" prediction of SIT: a system with macroscopic coherence should exhibit a stronger gravitational pull than an incoherent system of identical constituent mass. For example, a Bose-Einstein Condensate (BEC), where $R_{\text{coh}} \to 1$ globally, effectively creates a deeper well in the Time-Density field than a thermal gas of the same atoms. This anomaly, while negligible at the scale of a single atom, becomes a testable prediction for precision gravimetry experiments involving macroscopic quantum states.

6 Conclusion

This paper has presented a first-principles derivation of the Hydrogen atom's structure based entirely on the postulates of Super Information Theory. By shifting the ontological foundation of physics from mass and charge to Coherence and Time-Density, we have successfully recovered the two essential features of atomic mechanics: the inverse-square force law and the quantization of orbital energy levels.

6.1 Summary of Results

First, we applied the SIT Master Energy-Density Equation, $E = \zeta R_{\text{coh}}\rho_t^2$, to the radial gradient of the Time-Density field surrounding a nucleus. We demonstrated that the gradient of the squared field, $\nabla(\rho_t^2)$, naturally produces an attractive force scaling as $1/r^2$. This establishes that the "Coulomb" interaction is not a fundamental gauge force, but an entropic force arising from the geometry of the causal update rate.

Second, we addressed orbital stability through the lens of topological constraints. We established that the electron's mass is an emergent property contingent upon maintaining a non-zero Coherence Ratio ($R_{\text{coh}} > 0$). This imposes a strict condition on the orbital trajectory: the holonomy of the coherence bundle must be Identity. This topological requirement restricts the electron to discrete radii where the phase winding number is an integer ($\oint \nabla\theta \cdot dl = 2\pi n$), thereby deriving the Rydberg levels as geometric necessities rather than arbitrary postulates.

6.2 Final Remarks

The implications of this derivation extend far beyond the Hydrogen atom. It suggests that the "laws" of the Standard Model are emergent behaviors of a deeper informational substrate. The electron does not calculate eigenvalues of a differential operator; it simply follows the path of Least Informational Mismatch, existing only where its coherence can survive the destructive interference of the vacuum.

By successfully deriving the structure of matter from the dynamics of Coherence (R_{coh}) and Time-Density (ρ_t), Super Information Theory fulfills the promise of *Informational Realism*: proving that the solid reality we observe is, at its core, a structured resonance of information.

References

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